Engineering Chemistry · Unit_02 · Electrochemistry and corrosion

DRAFT — UE26CY151A · Unit_02 · Electrochemistry and corrosion · Types of corrosion - stress corrosion - Part III

Standalone video lesson · authored study notes with source citations

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· How to use this pack

Read the topic explanation, worked examples, and recap. Evidence citations point to source transcript intervals and slide pages; full slide images remain in the source visual index. Topics include the applicable parts: overview, key terms, formulas, figures, flowcharts, important points, worked examples, past-paper questions and a recap. IDs are shared with the separate documents: E = formula, G = graph, F = flowchart, W = worked example. Each ID is numbered by topic, so E4.2 is the second formula of topic 4.

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M1

Types of corrosion - stress corrosion - Part III

1 Types of corrosion - stress corrosion - Part III

Study focus: Types of corrosion - stress corrosion - Part III. This topic explains the central idea, its conditions, and how the lecture applies it in context. Review the selected source statements and follow their citations for the full lesson segment. EV0000158 · EV0000159

SHORT-NOTE RECAP

Core explanation · 00:00–04:59

We will continue with module six on corrosion chemistry.

Key details

- Then we move on to studying some of the factors which affect the rate of corrosion.
- Stress corrosion is also referred to as stress corrosion cracking, and this is a most common type of corrosion which happens with metals when they are under stress.
- and at the same time they are exposed to a specific corrosive environment.
- We know that whenever.
- We are doing any manufacturing or fabrication of articles, structures, etc., the metal is subjected to lot of mechanical operations such as pressing, hammering, rolling, bending, etc.
- Now, because of these operations on the metal to get the right structure there will be some a residual stress left inside the metal.
- Apart from that, that there could also be some external stress due to the service conditions.
- Give the right corrosive environment.
- We have an iron article.
- Is so much here, then a small crack is going to be initiated for inside the metal in order to relieve the stress.
- the bend which is here, the bent article over here.
- Now this being at higher energy level, they will be more reactive as compared to this part, and therefore the reactive part will become the anode and the stress free part becomes the cathode.
- We have to see how the anode and cathode are getting generated, and in this stress corrosion, this is how it is being formed wherever there is stress in the metal.

- Exposed to a specific corrosive environment, we can see that stress corrosion cracking is going to take place.
- We just now saw that there could be tensile stress in the metal but apart from that the specific corrosive environment is what is required.
- Just having the specific corrosive environment or the tensile strength is tr stress is not.
- Yeah sufficient both these factors the tensile stress as well as the specific corrosive environment when they act upon this metal, then the metal undergoes stress corrosion.
- The stressed part and the other part is acting as the cathode.
- This crack when we subject it to a corrosive environment we will find that due to corrosion this part being anode undergoes corrosion and we know what happens in corrosion the metal is going to get converted into iron.
- It's going to deepen and by the time it reaches here it has caused damage or breakdown of the structure.

Example or application from the lecture

- First, let us look at what is stress corrosion.
- Suppose we have a metal which is bent like this and a small craitiated in it.

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EV0000158 · EV0000159 · EV0000160 · +1 source units in pack.json

Core explanation · 05:00–09:59

But what we have to remember is both the conditions should be satisfied.

Key details

- The tensile stress as well as specific corrosive environment needs to be present for such a corrosion to take place.
- If the metal is break brass or the structure is made of brass, which is an alloy, this will be corroded if there is stress in the structure.
- This will get corroded.
- only in the presence of ammoniacal solution or ammoniacal vapor.
- cracking to take place.
- So therefore in the presence of both these conditions we will find that stress corrosion can happen and one more.
- Fact about stress corrosion is that it causes serious damage as we saw this crack once it starts deepening due to the corrosive environment, it can cause breakdown of structures and it has in the past causes caused air crashes, bridge collapses, and even border.
- Just now I mentioned that boiler explosions could happen due to this stress corrosion.
- Basically it's a form of stress corrosion and it takes place in boilers and we know that in what boilers what are we doing?
- We are heating water.
- If this comes in constant contact with concentrated alkali, then there is stress in the boiler and also you provided the specific corrosive environment, stress corrosion is going to happen.
- we get sodium hydroxide.

- And if we remember that for mild steel, what is the corrosive environment we require sodium hydroxide.
- there now what happens is because we are heating the boiler, it is at a high temperature, this sodium hydroxide which went inside the crack.
- is slowly getting more and more concentrated.
- When it gets concentrated to about 10%, then this sodium hydroxide starts dissolving all the iron with which it is in contact.

Example or application from the lecture

- Let us talk about it.
- Let us talk of how distress corrosion takes place in boilers.
- Let us see what exactly is happening.

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Core explanation · 10:00–14:59

And when it dissolves it forms a compound called sodium ferroate.

Key details

- After this, the sodium ferroate undergoes further hydrolysis, and when it undergoes further hydrolysis, it is going to look at this reaction sodium fydrohy ferro weight is undergoing hydrolysis giving you Fe three O four plus H two plus NOH.
- Again you have regenerated the sodium hydroxide.
- So that means throughout the you while you are using the boiler continuously you have sodium hydroxide in the water and apart from that we also got Fe three O four and Fe three O four is known to be brittle in nature.
- The iron which was around this crack which was initially gener initiated due to the due to the stress in the structure.
- Got dissolved and now it is forming something like Fe 3O4, which is brittle in nature.
- The crack now starts deepening.
- Because it is happening in the presence of sodium hydrogen.
- hydroxide and it is in a very specific environment.
- the mild steel is the water boiler made of mild steel is undergoing corrosion.
- Plus we are making the iron into something brittle like FE3O4 I mentioned just now.
- Therefore, this is called as this is known as caustic embrittlement.
- So because of this causing embrittlement, we get a corrosion cell.
- Whenever we talk of electrochemical corrosion, we know there should be a formation of anode and cathode.
- We'll always follow the same rule.
- We'll write the anode first and then the cathode.
- And it is in contact with concentrated sodium hydrox, whereas the stress free iron is going to act as the cathode, and that is in contact with the dilute sodium hydroxide.

- And that is the reason why electrochemical corrosion takes place, the stress as well as the specific environment in this case we have provided the sodium hydroxide.
- Now of course we have to think of ways by which this can be prevented because it could be very dangerous.
- Then we have the difference in electrode potentials of anode and cathode, nature of corrosion product, ratio of the anode to cathode area, then we have overvoltage.
- Polarization, pH, temperature, conductivity of the medium and humidity.
- The first one, nature of the metal.
- How much corrosion is going to happen is going to depend on how reactive the metal is.
- If it is a noble metal, we don't expect much reactivity.
- Therefore corrosion is also very limited.
- If it's very noble, then you hardly see any corrosion.
- Those metals which are very reactive we find that they will be undergoing corrosion.
- In fact, they are more susceptible to corrosion.
- So we can say that the ones with lower electrode potential, that is where the reduction potentials are negative, that means they want to undergo oxidation, and we know that corrosion is an oxidation process.

Example or application from the lecture

- Let us take up some of them.

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Core explanation · 15:00–19:59

Negative reduction potentials will be more reactive than those which have positive reduction potential values.

Key details

- They all have very negative values, and we know that they are highly susceptible to corrosion.
- On the other hand, we have noble metals like silver, gold, and platinum which have positive values and we find that they are less susceptible to.
- And that's exactly what we found the ones which are lying high above in the series will undergo corrosion.
- The ones which are very low in the series with positive values do not undergo corrosion.
- But there are exceptions like chromium, aluminium, titanium.
- Because they become passive.
- Now that means electrochemical series is not sufficient for us to give us the order in which corrosion will happen.
- Therefore, a new series is required, which is known as the galvanic series.
- And what does this galvanic series do?
- It is going to arrange the metal.

- And now we have an advantage, it also can arrange the alloys in the order of their tendency to undergo corrosion and in a particular environment because we know that corrosion depends not only on the metal or the alloy, it also depends on the environment to which the metal is subjected.
- Yes, to a good extent we can say that the electrochemical series gives us an order in which the metals will show tendency to undergo corrosion, but nevertheless, electrochemical series are standard values.
- That means we are using standard solutions and standard conditions, and moreover.
- Aluminium, you see here it has a very negative value.
- Aluminium, chromium also you'll see they have very negative values, but we know that aluminium is passive in nature.
- So that means electrochemical series is not sufficient.
- For us to decide the corrosion tendency.
- Tend and see not only metals, you can see some alloys also.
- All these are arranged in an order in which.
- They will undergo corrosion.
- and listed here which metal or alloy will be more reactive or undergo corrosion more as compared to the others.
- Always the environment is also listed over there.
- Coming to the next.
- reason or the factor which is going to affect the rate of corrosion and here the difference between the potential of the anode and the cathode region.
- We know that many times two metals are in contact with each other and.
- This is say M1 and this is M2.

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Core explanation · 20:00–24:59

We know also that the driving force here is the difference in the electrode potential.

Key details

- We are talking about galvanic corrosion.
- Higher is going to be this differential metal corrosion or galvanic corrosion.
- Because more is the potential difference, diff more is the potential difference, the free energy decrease accompanying the process is higher, and therefore we find that extent of corrosion also increases.
- But what we have to remember here, more the difference between the electrode potentials of the two metals, more is the anode which is going to undergo corrosion.
- It is a more reactive metal which undergoes corrosion.
- If we understand that this is the anode and this is the cathode, we will find more the difference between this anode and cathode.
- More is the corrosion of M1 metal in this case.
- So what is the first of all which metal will undergo corrosion?

- Iron, we can see very clearly.
- Because this will become the anode.
- And the potential difference here comes out to be 0.78 volts.
- What is the metal which is undergoing corrosion?
- Again we can see that it is iron which will undergo corrosion.
- Here also I am acting as anode, but now the potential difference has come down to 0.3 that is e cathode e0 cathode minus the e0 anode.
- Here it is 0.78, here it is 0.3.
- Therefore, though in both the cases it is iron which.
- We find that iron will undergo corrosion faster when it is in contact with copper.
- The use of this similar metal should be avoided wherever it is possible.
- Should be always made of the same metal, otherwise, we'll find that the metal which is more reactive.
- Will undergo corrosion.
- Because nuts and bolts are always in close contact, they should be made of the same metal.
- Now coming to the next very important factor which affects the rate of corrosion, nature of corrosion product.
- Now in some cases, just now we saw in the first factor that cr when we talked about nature of metal, chromium, aluminium, etc., they do not undergo corrosion.
- They become passive.
- Because once the corrosion product is formed on the surface of the metal, it acts as a protective film.
- This we should always remember.
- The metal, and now the corrosion product has got formed on the surface.
- product which is formed on the surface.
- So it's forming a nice film over the surface and protecting the metal how it is creating a barrier between the surface and the environment and therefore we will find that the metal is now cut off from the environment and hence there can be no further corrosion.
- So, if the corrosion product is insoluble, stable, uniform, and non-porous, we find that it forms a protective film and therefore prevents further corrosion.
- It acts as a barrier, a thin, invisible, impervious sometimes continuous film is formed, and this acts as a barrier between the fresh metal surface and the corrosion environment.
- Passive, and that's exactly what we see in the case of chromium and aluminium.

Example or application from the lecture

- Let us take up a case one where we have iron which is minus 0.44 volts and copper 0.34 volts.
- Let us take another case where we have iron at minus 0.44 and tin is the metal which is in close contact with it and that tin has an electrode potential standard electrode potential of minus 0.14.
- For example, nuts and bolts.

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Core explanation · 25:00–29:59

metals like iron, zinc and magnesium.

Key details

- A non-continuous film will be formed.
- Even though the corrosion product is formed, it is not protecting the surface underneath, and every time there is a new surface after the corrosion product is formed, there's a new surface which is getting exposed.
- and therefore the corrosion will continue to happen continuously.
- So metals like iron, zinc, and magnesium, the corrosion product is soluble, unstable, non-uniform, and porous, and therefore there is no stopping corrosion.
- It will keep on getting corroded as a new metal surface, a fresh metal surface is getting extra.
- Which is covering the surface, but it is soluble in the surrounding environment.
- getting exposed every time.
- that this will result in more and more corrosion of the metal.
- We all use stainless steel utensils for cooking and other work in the at homes.
- But we know that it does not get corroded.
- Because it has.
- It is an alloy which contains chromium.
- steel or that is iron it iron containing some amount of carbon so that steel articles we find that they get corroded easily like your nails or any steel article will get corroded but stainless steel due to the.
- Passive nature of chromium which has been added there.
- This chromium oxide forms a nice layer on the surface and forms a protective film and prevents further corrosion of the particle.
- Now in tomorrow's class in the next class we are going to talk about take up some more factors like overvoltage, polarization, pH temperature, etc, which were listed.
- It just now, and then we will learn how those factors are going to affect corrosion rate.

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A Symbol glossary

Symbol	Name	Meaning	Appears in

B Sources, coverage and gaps

Source	Path	Use
Types of corrosion - stress corrosion - Part III transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/AV_Summaries/008_Types_of_corrosion_-_stress_corrosion_-_Part_III__5da0c5dd14cf8164.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Types of corrosion - stress corrosion - Part III transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/AV_Summaries/008_Types_of_corrosion_-_stress_corrosion_-_Part_III__5da0c5dd14cf8164.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Types of corrosion - stress corrosion - Part III transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/AV_Summaries/008_Types_of_corrosion_-_stress_corrosion_-_Part_III__5da0c5dd14cf8164.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9992; only the JSON segmentation supplies the displayed timestamps.
Types of corrosion - stress corrosion - Part III transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/AV_Summaries/008_Types_of_corrosion_-_stress_corrosion_-_Part_III__5da0c5dd14cf8164.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Historical question-source excerpts are shown only when a page has a strong text-based unit match. They are source references, not new solutions or exam predictions.

Known gaps, conflicts and corrections

1. 1 transcript segments have low ASR confidence; they are flagged with timestamps in the transcript blocks and indexed in the corpus.
2. This is a visibly labelled source-preservation draft. Human page-level semantic and visual review is not complete; no unsupported subject summary or formula correction is claimed.
3. The 19.26 GiB MP4 files remain hosted on Hugging Face; the full video path/hash and transcript coverage are indexed. The full audio/video was not watched end-to-end.
4. Formula candidates are machine-extracted text with source locators; use the adjacent original slide page and the formula index before treating a candidate as an authoritative equation.

5. The study PDF contains edited, topic-organized lecture notes and textual slide-page takeaways. Full-page slide images remain in the source/visual index datasets, not embedded as the lesson. Human content, ASR, formula, and page-level visual review remains pending.

Coverage

Evidence is summarized by source to keep the reading copy legible. Full source-unit IDs, locators and decisions remain in pack.json.

Source	Unit counts	Example decisions
SRC00034	6 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0042. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0043 to s0062. Sidecar variants compare...
SRC00035	6 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0042. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0043 to s0062. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00036	6 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0042. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0043 to s0062. Sidecar variants compare...
SRC00037	6 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0042. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0043 to s0062. Sidecar variants compare...

C Revision and retrieval practice